

NCERT Solutions For Class 12 Computer Science

Chapter 12 – Security Aspects

Exercise Solutions

1. Why is a computer considered to be safe if it is not connected to a network or Internet?

A computer not connected to a network or Internet is considered safe because it is isolated from external threats such as malware, hacking attempts, and unauthorized access, which generally enter through networks or the Internet.

2. What is a computer virus? Name some computer viruses that were popular in recent years.

A computer virus is a piece of software code created to perform malicious activities and harm system resources like CPU time, memory, personal files, or sensitive data.

Some well-known viruses are CryptoLocker, ILOVEYOU, MyDoom, Sasser and Netsky, Slammer, and Stuxnet.

3. How is a computer worm different from a virus?

Unlike a virus, a worm is a standalone program and does not need a host program to spread. Worms self-replicate and propagate through networks automatically, while viruses require a user to execute an infected file for replication.

4. How is Ransomware used to extract money from users?

Ransomware is malicious software that blocks user access to their data or threatens to publish personal data unless a ransom is paid. It often demands payment in cryptocurrencies like Bitcoin to restore access.

5. How did a Trojan get its name?

A Trojan gets its name from the story of the Trojan Horse in Greek mythology. It is malware that looks like legitimate software and tricks the user into installing it, similar to how Greek soldiers hid inside a wooden horse to enter Troy.

6. How does an adware generate revenue for its creator?

Adware generates revenue by displaying online advertisements such as pop-ups, web pages, or installation screens. Creators earn money either by displaying these ads or through "pay per click" mechanisms when users interact with the advertisements.

7. Briefly explain two threats that may arise due to a keylogger installed on a computer.

Keyloggers can capture and leak sensitive information such as passwords and bank account details.

They may record private communications and personal data, leading to privacy breaches and potential identity theft.

8. How is a Virtual Keyboard safer than On Screen Keyboard?

An on-screen keyboard has a fixed QWERTY layout, making it easier for keyloggers to record key presses. Virtual keyboards randomise the key layout each time, making it difficult for keyloggers to capture accurate input, thus providing greater safety for sensitive information.

10. List and briefly explain different modes of malware distribution.

- 1. Downloaded from the Internet:** Malware disguised as legitimate files is downloaded unknowingly by users.
- 2. Spam Email:** Malicious attachments or links in unsolicited emails introduce malware.
- 3. Removable Storage Devices:** Infected pen drives, memory cards, and mobile devices carry malware to other systems.
- 4. Network Propagation:** Some malware, like worms, spread automatically through network connections.

11. List some common signs of malware infection.

Frequent pop-up windows prompting downloads
Changes to browser homepage
Mass emails sent from your account
Slow computer and frequent crashes
Unknown or unwanted programs launching at startup
Files or programs appearing/disappearing suddenly

12. List some preventive measures against malware infection.

Use and regularly update antivirus and anti-malware software

Set browser security settings appropriately

Check for the lock icon on secured websites

Install only licensed and updated software

Take regular data backups

Enable firewall protection and avoid suspicious links/emails

Scan removable storage and don't share passwords

13. Write a short note on different methods of malware identification used by antivirus software.

1. Signature-based detection: Uses virus signature databases to detect known malware.

2. Sandbox detection: Runs files in a virtual environment to observe behaviour.

3. Data mining techniques: Uses machine learning to classify files.

4. Heuristics: Compares program code to known viruses for similarities.

5. Real-time protection: Monitors processes and applications during execution.

14. What are the risks associated with HTTP? How can we resolve these risks by using HTTPS?

HTTP sends data as plain text, making it easy for hackers to intercept and manipulate information. HTTPS solves this risk by encrypting data during transmission and using SSL certificates for secure communication between client and server.

15. List one advantage and disadvantage of using Cookies.

Advantage: Cookies store browsing details like preferences and login, making user experience smoother and more personalized.

Disadvantage: Some third-party cookies track and share user data for advertising without user consent, raising privacy concerns.

16. Write a short note on White, Black, and Grey Hat Hackers.

1. White Hats (Ethical Hackers): Use expertise to fix security flaws and help organizations.

2. Black Hats (Crackers): Exploit vulnerabilities for illegal purposes or disruption.

3. Grey Hats: Hack for fun or challenge; actions may not always be harmful or beneficial.

17. Differentiate between DoS and DDoS attack.

1. DoS (Denial of Service): Attack from a single source floods a resource, blocking legitimate users. Easier to resolve by blocking the attacker.

2. DDoS (Distributed Denial of Service): Multiple compromised systems (zombies) flood the target, making mitigation difficult as the attack comes from many sources.

18. How is Snooping different from Eavesdropping?

Snooping involves secretly collecting and analyzing network traffic, which can be stored for later review and is not always malicious (e.g., troubleshooting). Eavesdropping refers to real-time unauthorized interception of communication, such as live calls, and is always malicious in nature.